5.2 Logarithmic Functions and Their Graphs

Name _____ Period ____ Date _____

Pre-Calculus 5.2 · Student sheet

I can…

- rewrite between logarithmic and exponential form.
- evaluate logarithms and use their basic properties.
- graph logarithmic functions and find their domains.

Vocabulary · fill in the missing word

Logarithmic function — $y =$ _____ $= a^y$ $(a > 0, a \neq 1)$

Common logarithm — $\log x =$ _____

Natural logarithm — $\ln x =$ _____

Part 1 · Swap the columns

Swap the columns of $y = 2^x$. Plot both, with the line $y = x$.

$y = 2^x$: x	−2	−1	0	1	2	3
y	1/4	1/2	1	2	4	8
Swapped: x						
Swapped: y						

Part 2 · Notes and examples (fill in each note, then work the example)

Definition: $y = \log_a x$ if and only if _____.

Example 1

a. Write $\log_2 8 = 3$ and $\log_5(\frac{1}{25}) = -2$ in exponential form.

b. Write $4^3 = 64$ in logarithmic form.

Example 2

Evaluate without a calculator.

a. $\log_2 32$ b. $\log_3 1$ c. $\log_4 2$ d. $\log(\frac{1}{100})$

Properties: $\log_a 1 = 0$, $\log_a a = 1$, $\log_a a^x = x$, and a^($\log_a x$) = _____.

Example 3

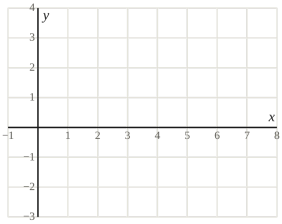
Simplify or solve.

a. $\log_7 7$ b. $6^{\log_6 20}$ c. $\ln e^5$ d. $\log_3 x = \log_3 12$

Graph of $y = \log_a x$: x-intercept (1, 0), vertical asymptote _____, domain $x > 0$.

Example 4

Graph $f(x) = \log_3 x$ and $g(x) = \log_3(x - 2) + 1$. Give each asymptote and x-intercept.



Domain: the expression inside a logarithm must be _____.

Example 5

Find each domain.

a. $f(x) = \ln(x - 2)$

b. $g(x) = \ln(2 - x)$

c. $h(x) = \ln x^2$

Part 3 · Partner practice

Partner 1

Evaluate $\log_2(\frac{1}{8})$ and $\log 1000$.

Partner 2

Find the domain and vertical asymptote of $f(x) = \log_4(x + 5)$.